

Why human lifespan is rapidly increasing: solving "longevity riddle" with "revealed-slow-aging" hypothesis



Healthy life span is rapidly increasing and human aging seems to be postponed. As recently exclaimed in Nature, these findings are so perplexing that they can be dubbed the 'longevity riddle'. To explain current increase in longevity, I discuss that certain genetic variants such as hyper-active mTOR (mTarget of Rapamycin) may increase survival early in life at the expense of accelerated aging. In other words, robustness and fast aging may be associated and slow-aging individuals died prematurely in the past. Therefore, until recently, mostly fast-aging individuals managed to survive into old age. The progress of civilization (especially 60 years ago) allowed slow-aging individuals to survive until old age, emerging as healthy centenarians now. I discuss why slow aging is manifested as postponed (healthy) aging, why the rate of deterioration is independent from aging and also entertain hypothetical use of rapamycin in different eras as well as the future of human longevity.